



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
20-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A spherical raindrop of initial mass M_0 and initial radius R falls freely from rest. Let ρ be the constant density and r the radius at time t . Assume that the drop maintains its spherical shape. The drop loses mass at the rate $\frac{dm}{dt} = -4\pi k \rho r^2$, where k is a constant. Define $t_c = \frac{R}{k}$. Select the correct statement (s) :

(A) The speed of the drop at time $\frac{t_c}{2}$ is $\frac{15gR}{8k}$.

(B) The rate of mass loss goes to zero as time $t \rightarrow t_c$

(C) The drop loses all its mass in time t_c

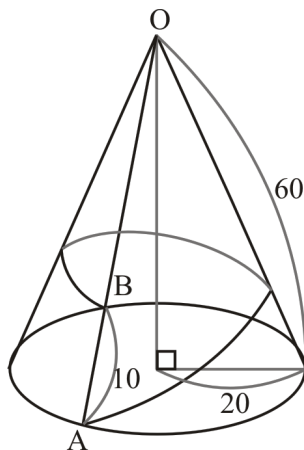
(D) The momentum approaches $\frac{M_0 g R}{4k}$ as $t \rightarrow t_c$

Space for Rough Work

2. A particle executes simple harmonic motion with time period T , amplitude A and maximum speed V_m . The particle is at the mean position when $t = 0$. If y is the displacement from the mean position and v is the speed, then choose the correct statement (s) from the following in the time interval $0 \leq t \leq \frac{T}{4}$.
- (A) For $v = \frac{V_m}{2}$, $y > \frac{A}{2}$
- (B) At $y = \frac{A}{2}$, the kinetic and potential energies are equal.
- (C) The magnitude of the acceleration at $y = \frac{A}{2}$ is twice that at $y = \frac{A}{4}$.
- (D) At $t = \frac{T}{8}$, $y > \frac{A}{2}$
3. A particle performs uniform circular motion of radius R and speed v about the origin O . Then the angular speed of the particle about :-
- (A) O is $\frac{v}{R}$.
- (B) a point close to the circumference and inside the circle exhibits a sharp maximum.
- (C) a point on the circumference is $\frac{2v}{R}$.
- (D) a point outside the circle is zero at two instants during the motion of the particle around the circle.

Space for Rough Work

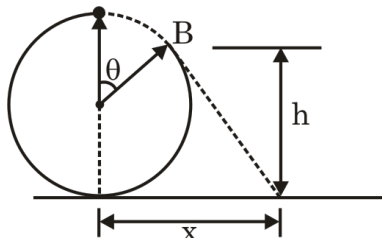
4. The diagram illustrates a right circular cone-shaped hypothetical mountain, of apex "O". If you build a shortest distance track for a sightseeing train which goes exactly once around the mountain, in which the track starts at point A and ends at point B. the track will first go uphill, but then it will go downhill. A & B will lie on the same slant line from A to apex O such that distance between them along the slant line, $AB = 10$ km. radius of cone = 20 km. Slant length of the cone = 60 km. The speed of train (Consider it a particle) is set at a constant 20 km/h. Mark the correct statement(s).



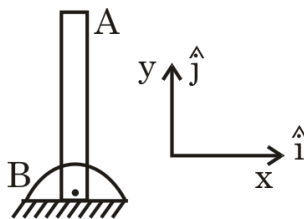
- (A) At the instant of shortest distance from "O" the angular velocity of train about "O" is maximum.
- (B) The journey time of the train is $\frac{\sqrt{91}}{2}$ hr.
- (C) The length of the downhill part of the journey is $\frac{200}{\sqrt{91}}$ km.
- (D) The velocity of the train is always perpendicular to the line joining O to the train.

Space for Rough Work

5. A particle with mass m is at rest on top of a smooth massive infinitely long cylinder lying on a horizontal plane. The radius of the cylinder is r . A small push is applied to the particle to get it sliding down the cylindrical surface. The particle leaves the surface at a point B on the cylinder which is at a height h and reaches the ground at a distance x from the cylinder. A cross-sectional view is shown in the figure. The radius vector connecting the point B makes an angle θ with the vertical. Then :-



- (A) h is independent of m and depends on r .
 (B) x is independent of m and r .
 (C) the speed of the particle at point B is independent of m and depends on r .
 (D) the angle θ is greater than 45° .
6. A uniform rod of mass 1 kg and length $L = 1\text{ m}$ stands vertical. Rod is free to rotate about hinge at B in x - y plane only. A force $\vec{F} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ starts acting on the rod at point A. Then immediately after force $\vec{F}$ starts acting on the rod



- (A) angular acceleration of rod will be $9(-\hat{k}) \text{ rad/s}^2$
 (B) angular acceleration of rod will be $12\sqrt{34}(-\hat{k}) \text{ rad/s}^2$.
 (C) reaction torque of hinge on rod about B will be $5(-\hat{i}) \text{ Nm}$.
 (D) reaction torque of hinge on rod about B will be 0.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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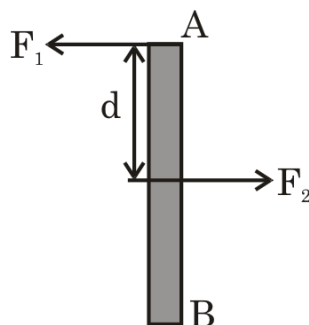
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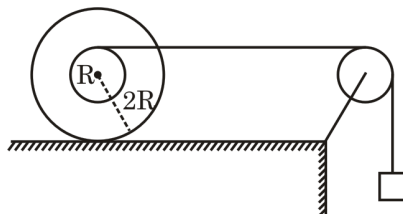
-
1. A thin rod of mass 0.3 kg and length 0.2 m is suspended horizontally by a metal wire which passes through its centre of mass and is perpendicular to its length. The rod is set to torsional oscillation with a period of 2s. The thin rod is replaced by an equilateral triangular lamina which is suspended horizontally from its centre of mass. If its period of oscillations is found to be 10s, the moment of inertia of the triangular lamina about the axis of suspension is approximately $\alpha \times 10^{-2} \text{ kg m}^2$. Find the value of α .
 2. A particle of mass m moves under the influence of the potential energy $V(x) = P/x^2 - Q/x$. Here P, Q are real positive constants. The angular frequency of small oscillations of the particle around the equilibrium point is (in rad/s) [Take $Q = 14 \text{ Jm}$, $m = 0.5 \text{ kg}$, $P = 4 \text{ Jm}^2$]
 3. A mass of 1kg is dropped from a height of 2m on a horizontal spring board. The vertical spring supporting the board has a spring constant of 87.5 N/m. The maximum distance (in m) by which the mass compresses the spring is:
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Space for Rough Work

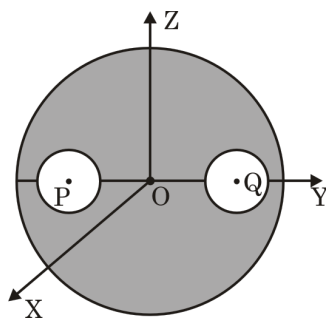
4. A thin uniform rod of mass 1 kg moves translationally with acceleration $a = 2 \text{ ms}^{-2}$ due to two antiparallel forces $\vec{F}_1$ and $\vec{F}_2$ as shown. The distance between the points at which these forces are applied is equal to $d = 20 \text{ cm}$. If $F_2 = 5 \text{ N}$, length of the rod (in m) is:



5. In the figure shown, the spherical body and block, each have a mass m . The moment of inertia of the spherical body about centre of mass is $2mR^2$. The spherical body rolls without slipping on the horizontal surface. The ratio of kinetic energy of the spherical body to that of the block is :



6. A solid sphere of uniform density and mass M has radius 4m . Its centre is at the origin of the coordinate system. Two spheres of radii 1m are taken out so that their centres are at $P(0, -2, 0)$ and $Q(0, 2, 0)$, respectively. This leaves two spherical cavities. What is the gravitational field at the origin of the coordinate axes (in multiples of GM) ?



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

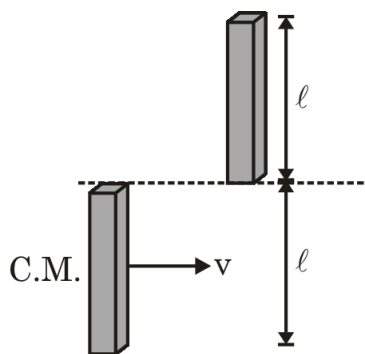
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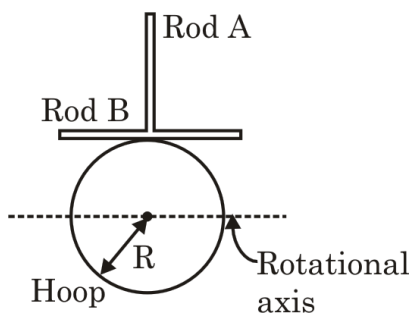
1. A bar of mass m , length ℓ is in pure translatory motion with its centre of mass velocity v . It collides with and sticks to another identical bar at rest as shown in figure. Assuming that after collision it becomes one composite bar of length 2ℓ , the angular velocity of the composite bar is $\frac{kv}{4\ell}$ in anticlockwise direction. Find the value of k .



2. A carrom coin of radius 1.5 cm and mass 5g strikes the side of the board at an angle 30° with respect to the normal. The coin stays in contact with the side for 10^{-3} s and rebounds at an angle of 45° with respect to the normal. The initial speed of the coin is 1m/s and its final speed is 0.7 m/s. The average impulsive force (in N) on the coin in the normal direction is (round off to nearest integer)

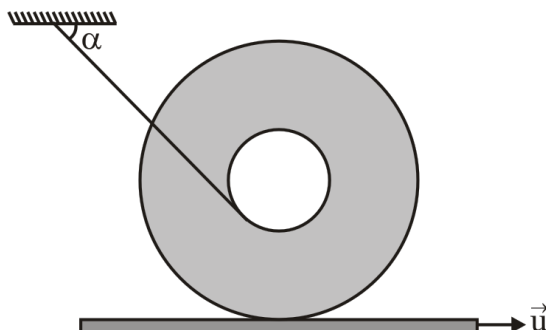
Space for Rough Work

3. A girl of mass 50 kg jumps from the shore onto a stationary boat of equal mass floating on a still pond. She jumps back to the shore almost immediately. Her speed in both cases is 2 m/s with respect to the shore. Consider her entire motion to be horizontal and in one dimension. Ignore friction and drag forces. The speed of the boat (in m/s) with respect to the shore after the jumps back is:
4. A rigid sculpture, consisting of thin hoop (of mass m and radius R) and two thin rods (each of mass m and length $L = 2R$), is arranged as shown in the figure. The sculpture is pivoted and can rotate about a horizontal axis in the plane of the hoop and passing through centre of the hoop. The sculpture's rotational inertia about this axis is approximately KmR^2 . Find the value of K . (Round off to nearest integer)



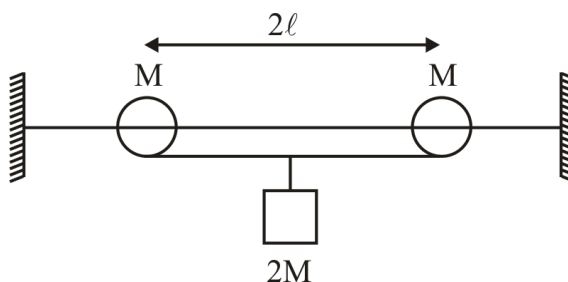
Space for Rough Work

5. A spool is placed at the platform, which moves in horizontal direction with velocity u . What should be speed of the center of the spool v_0 (in m/s) for its motion with the platform without slippage, if non-stretchable thread bound at the inner cylinder is always under tension? Solve the problem for the case when the thread makes angle α with horizon and inner radius of the spool is half of its outer radius. ($u = 0.6$ m/s, $\alpha = 37^\circ$.)



6. On a smooth horizontal rod, two beads of mass M each are connected by an inextensible thread of length 2ℓ as shown. A mass $2M$ hangs from the middle of the string as shown. Initially the string is horizontal when hanging mass is released. What is the maximum speed of $2M$ during the course of motion?

Take $\ell = \frac{15}{4} \sqrt{\frac{3}{2}}$ m, $g = 10$ m/s². (Round off to nearest integer.)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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-
1. Which of the following is not correct reason for controlled roasting for extraction of Sn from cassiterite.
- (A) For complete conversion of sulphides into their oxide
- (B) For conversion of impurity sulphide into their soluble sulphate which should not be further decomposed to produce their oxide
- (C) For conversion of metal oxides to silicates
- (D) For conversion of sulphide of metal to be extracted into its soluble sulphate
-

Space for Rough Work

2. Which of the following statement is correct for extraction of lead.
- (A) Carbon reduction is carried out for low grade ore
- (B) Self reduction is carried out for high grade ore
- (C) H_2SiF_6 is added into electrolyte in its electrorefining
- (D) Cu_2S remains unaffected during roasting due to high oxygen affinity of Fe
3. Incorrect option is/are.
- (i) Cu-salt gives red transparent bead under cold reducing flame in borax bead test
- (ii) K^+ gives violet/lilac flame when Na^+ is present as impurity with it
- (iii) Thenard blue confirms Zn-compound
- (iv) Cr-compound produces yellow bead in sodium carbonate bead test having formula of Na_2CrO_4
- (A) (i) only
- (B) (ii) & (iv) only
- (C) (ii) & (iii) only
- (D) (i) (ii) & (iii) only
4. $\text{M}^{n+} + \text{Na}_2\text{HPO}_4 \rightarrow \text{yellow ppt.}$
 M^{n+} is/are :-
- (A) Al^{3+}
- (B) Ag^+
- (C) Mn^{2+}
- (D) Fe^{3+}

Space for Rough Work

5. $\text{HNO}_3 + \text{As}_2\text{O}_3 \rightarrow \text{H}_3\text{AsO}_4 + \text{compound 'M'}$.
Which is/are incorrect regarding 'M'?
- (A) It may exist in two form
(B) It is the anhydride of HNO_3
(C) In both form N-atoms have same oxidation state
(D) The solid 'M' consists of ion pair structure
6. Which of the following is correct regarding perchlorates.
- (A) HClO_4 is the strongest acid known
(B) MgClO_4 is used as the electrolyte in so called dry batteries
(C) NH_4ClO_4 is known as anhydron
(D) RbClO_4 and CsClO_4 are insoluble in H_2O

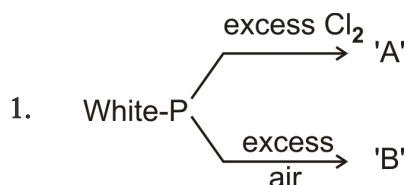
Space for Rough Work

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If number of 90° angles in A = x

and if the number of P-O bond in B = y

Then find the value of $\left(y + \frac{1}{x}\right)$:

2. Consider the species $\text{S}_9\text{O}_6^{2-}$
If the number of linkages between same element is x and minimum number of equivalent bonds is y, then find the value of x/y.
3. $\text{CH}_3\text{CO}_2\text{Na(s)} + \text{As}_2\text{O}_3\text{(s)} \xrightarrow{\Delta} \text{fumes having nonseating smell} + \text{CO}_2 + \text{Na}_2\text{CO}_3$
Find the ratio between the number of σ -bonds and number of carbon atom in one molecule of above unknown fumes.

Space for Rough Work

4. Find the ratio of number of σ -bond to number of π -bonds in one molecule of methylene blue.
5. Find ratio between total number of atoms in one molecule of colemanite and oxidation state of element whose oxide is an glass forming oxide.
6. $[\text{Au}(\text{CN})_2]^- + \text{Zn} \rightarrow \text{soluble complex (A)} + \text{Au}\downarrow$
If the number of electrons present in penultimate shell in Zn is 'a'
and if the maximum number of atoms present in soluble complex (A) is b,
Then find the value of (a/b).
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Space for Rough Work

SECTION-III : (Maximum Marks: 18)

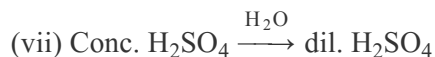
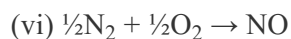
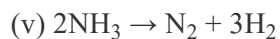
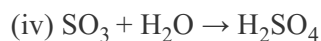
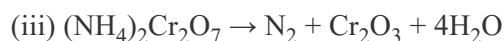
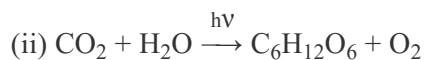
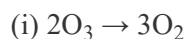
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Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. How many of the following reactions is/are endothermic.



2. $\text{S}_8 + \text{XeF}_2/\text{HF} \rightarrow \text{Xe} + \text{'A'}$

The number of bonds of identical length is _____.

3. How many of the following compounds are blue coloured?

Bi_2S_3 , $\text{C}_6\text{H}_5\text{N} \rightarrow \text{CrO}_5$, $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$, Malachite, Al_2S_3 , $\text{Cu}_3[\text{Fe}(\text{CN})_6]_2$, $\text{CoCl}_2(\text{anhydrous})$,
 $\text{Co}[\text{AlO}_2]_2$, NiS , $\text{Co}(\text{BO}_2)_2$, Mo_2O_5

Space for Rough Work

4. How many of the following reagent gives coloured ppt with AgNO_3 , which are soluble in NH_4OH .
- (i) Na_2SO_3
 - (ii) Na_2S
 - (iii) NaCl
 - (iv) Na_2CrO_4
 - (v) NaI
 - (vi) HCO_2Na
 - (vii) $\text{Na}_2\text{S}_2\text{O}_3$
 - (viii) NaNO_2
 - (ix) $\text{K}_2\text{Cr}_2\text{O}_7$
 - (x) $\text{CH}_3\text{CO}_2\text{Na}$
5. How many of the following processes/compounds is/are related to Al-extraction from red bauxite.
- (i) CO
 - (ii) Coke powder
 - (iii) $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
 - (iv) CaF_2
 - (v) $\text{Al}(\text{OH})_3$
 - (vi) NaOH
 - (vii) CO_2
 - (viii) BaF_2
 - (ix) Graphite
6. Find the number of incorrect match from the following :
- (i) Leaching : Mc-Arthur Forest Process
 - (ii) $\text{M} \rightarrow \text{M-oxide}$ curve : (+)ve slope in E. diagram
 - (iii) Amalgam : One component must be Ag
 - (iv) Slag : Presents the reoxidation of metal
 - (v) Flux : Impurity present in the ore
 - (vi) Illemanite : Ore of Mn
 - (vii) Green flame on the bassemer converter : Indicates completion of slag formation
-

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Sum of the series $S = \sum_{r=0}^n \frac{3^{r+4} \binom{n}{r}}{r+4 C_4} + \sum_{r=0}^3 \frac{n+4 C_r 3^r}{n+4 C_4}$ is

(A) $\frac{4^{n+4}}{n+3 C_n + n+3 C_{n-1}}$

(B) $4^{n+4} \binom{2n}{C_4}$

(C) $\frac{4^{n+4}}{n+4 C_4}$

(D) $\frac{3^{n+4} + 2^{n+4}}{n+4 C_4}$

2. Let there are n planes in R^3 such that any three have exactly one point in common and no four of them have a point in common. If $f(n)$ represent the number of parts in which these n planes will divide the space, then-

(A) $f(3) = 8$

(B) $f(4) = 15$

(C) $f(4) = 12$

(D) $f(10) = 176$

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3. Let us consider the matrices, defined as

$$A = \begin{vmatrix} \sin \theta & \cos \theta & \tan \frac{\pi}{11} \\ \sin\left(\frac{4\pi}{3} + \theta\right) & \cos\left(\frac{4\pi}{3} + \theta\right) & \tan \frac{2\pi}{11} \\ \sin\left(\frac{8\pi}{3} + \theta\right) & \cos\left(\frac{8\pi}{3} + \theta\right) & \tan \frac{3\pi}{11} \end{vmatrix}, B = \begin{vmatrix} \sin \theta & \cos \theta & \tan \frac{\pi}{11} \\ \sin\left(\frac{8\pi}{3} + \theta\right) & \cos\left(\frac{8\pi}{3} + \theta\right) & \tan \frac{2\pi}{11} \\ \sin\left(\frac{4\pi}{3} + \theta\right) & \cos\left(\frac{4\pi}{3} + \theta\right) & \tan \frac{3\pi}{11} \end{vmatrix} \quad \&$$

$$C = \begin{vmatrix} \sin\left(\frac{4\pi}{3} + \theta\right) & \cos\left(\frac{4\pi}{3} + \theta\right) & \tan \frac{\pi}{11} \\ \sin \theta & \cos \theta & \tan \frac{2\pi}{11} \\ \sin\left(\frac{8\pi}{3} + \theta\right) & \cos\left(\frac{8\pi}{3} + \theta\right) & \tan \frac{3\pi}{11} \end{vmatrix} \text{ then}$$

- (A) $\det(A) + \det(B) = 0$ (B) $\det(B) + \det(C) = 0$
 (C) $\det(A) + \det(C) - \det(B) > 0$ (D) $\det(A) + \det(B) - \det(C) > 0$
4. Let M, N are two non-singular matrix of order 3 with real entries such that $\text{adj}(M) = 2N$ and $\text{adj}(N) = M$, then -
- (A) $\text{adj}(M^2N) + \text{adj}(MN^2) = 4(M + 2N)$ (B) $M = 2N^{-1}$
 (C) $MN = 4I$ (D) $\text{adj}(MN^{-1}) = 4M^{-2}$

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5. Which of the following is/are correct ?
- (A) Number of words formed by letters of "ENGINEER" such that vowels occupy odd position are 48
- (B) Number of words formed by letters of "ENGINEER" such that G and R both appear before I are 1120
- (C) Number of words formed by letters of "CALCULUS" such that 2C's are separated and 2L's are together are 900
- (D) Number of words formed by letters of "CALCULUS" such that 2C's and 2L's are separated are 2830

6. If $(x + 1)(x + 2) \dots (x + 99) = A_0 + A_1 \cdot x + A_2 \cdot x^2 + \dots + A_{99} \cdot x^{99}$ and

$$S = \frac{\sum_{k=1}^{99} (k \cdot A_k)}{100!} = \sum_{i=1}^{99} \left(\frac{1}{r_i} \right), (r_i < r_j, \forall i < j) \text{ then value of } \sum_{k=1}^{99} \frac{1}{\sqrt{(r_k - 1)}} \text{ is lying between}$$

- (A) 15 and 19 (B) 15 and 17 (C) 14 and 17 (D) 17 and 19

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $1 \cdot ({}^{20}C_0)^2 + 2 \cdot ({}^{20}C_2)^2 + 3 \cdot ({}^{20}C_4)^2 + \dots + 11 \cdot ({}^{20}C_{20})^2$ is equal to $3 [{}^{8a}C_{4a} + {}^{4a}C_{2a}]$, then a is equal to
2. Let $A = \begin{bmatrix} 4 & a \\ -1 & b \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ be two matrices satisfying the relation, $A^3 + 3A^2B + 3AB^2 + B^3 = (A + B)^3$. If $(A + B) = \lambda(A^{-1} + B^{-1})$, then λ is equal to -
3. The number of ways in which 15 boys and 2 girls can sit in a row such that between the girls at the most 2 boys sit is 'k' then $\left(\frac{k}{15! 5!} \right)$ is equal to

Space for Rough Work

4. Let $f(x) = \sum_{r=0}^n \sum_{k=r}^n {}^k C_r$ then total number of divisors of $f(9)$ is equal to :
5. The number of 6-digit numbers whose digits are selected from set $\{5, 6, 7, 8, 9\}$ such that any digit that appears in the number appears at least twice is :
6. $\left[\frac{2}{1} \cdot {}^7 C_0 (2)^0 (3)^7 - \frac{3}{2} \cdot {}^7 C_1 (2)^1 (3)^6 + \frac{4}{3} \cdot {}^7 C_2 (2)^2 (3)^5 - \dots - \frac{9}{8} \cdot {}^7 C_7 (2)^7 \right] - 410$ is equal to
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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Consider invertible matrix $M = \begin{bmatrix} x & 0 & x \\ 3 & 1 & 2 \\ -x & 2 & x \end{bmatrix}$. If determinant value of adjoint matrix of matrix M is k for exactly two distinct real values of x, then minimum integral value of '9k' is-
2. Let there be 15 letters $\left(L, M, N, O, P, \underbrace{Q, Q, Q, \dots, Q}_{10 \text{ Q's}} \right)$. If the arrangement of these letters in a line so that there is atleast two alike letters (Q) between two distinct letters is $\frac{k!}{2}$, then 'k' is equal to
3. The value of $m^m - \sum_{r=1}^m \frac{(m+1)(r-1)m^{r-1}}{r^m C_r}$ is equal to _____

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4. Consider the system of equations

$$2x + py + 4z = 6$$

$$x + 2y + qz = 5$$

$$x - y + 2z = 3$$

If system has a unique solution (α, β, γ) and $\alpha + \beta + \gamma = 1$, then q is-

5. If L.C.M. of A, B, C is $r^2 t^2 s^2$, where r, s, t are distinct prime numbers and A, B, C are positive integers, the number of ordered triplets (A, B, C) is K , then value of $\left[\frac{K}{1000} \right]$ is (where $[.]$ denotes greatest integer function)

6. If A, B, C are angles of an acute angle triangle. The minimum value of

$$\frac{1}{3^5} \begin{vmatrix} (\tan B + \tan C)^2 & \tan^2 A & \tan^2 A \\ \tan^2 B & (\tan C + \tan A)^2 & \tan^2 B \\ \tan^2 C & \tan^2 C & (\tan A + \tan B)^2 \end{vmatrix} \text{ is}$$

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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